Independent University, Bangladesh

CSE313 Compiler Construction · Sample Midterm + Quiz (Practice) · Time: 1h30m

Same shape as Mid Autumn-23. New questions.

Question paper

Mid Question (Total marks: 20)

1. Prove whether the following grammar is Ambiguous or NOT. If the grammar is ambiguous, convert it to unambiguous form. **[7]**

```
E → E - E | E / E | E ^ E | ( E ) | id
```

2. Write the First() and Follow() of the following grammar. Draw the LL(1) Parsing Table and parse the string `w x x y k z` showing all steps. If parsing is not possible, point out the reasons. (b) Using the same table, parse `x y w z`. **[8]**

```
S → X Y z | w S
X → x X | ε
Y → y Z | ε
Z → k | ε
```

3. Find True/False: **[5]**

(a) An LL(1) grammar can be left recursive.
(b) The FIRST set of a non-terminal can contain ε.
(c) A grammar with a FIRST/FIRST conflict can be parsed by an LL(1) parser.
(d) The lexical analyzer removes comments and white space.
(e) Left factoring changes the language of the grammar.

Quiz Question (Total marks: 10)

1. Convert the following Non-Deterministic Grammar (NDG) to Deterministic Grammar (DG): **[5]**

```
S → a b c S | a b d | a b | a T | e
T → t
```

2. Convert the following Left Recursive Grammar (LRG) to Right Recursive Grammar (RRG): **[5]**

```
A → A b c | A d | e | f B
B → B g | h
```

Extra practice [8]

4. The grammar below is left recursive. Make it ready for LL(1), then write First() and Follow(), draw the LL(1) table and parse `int id ; float id`. **[8]**

```
L → L ; D | D
D → int id | float id
```